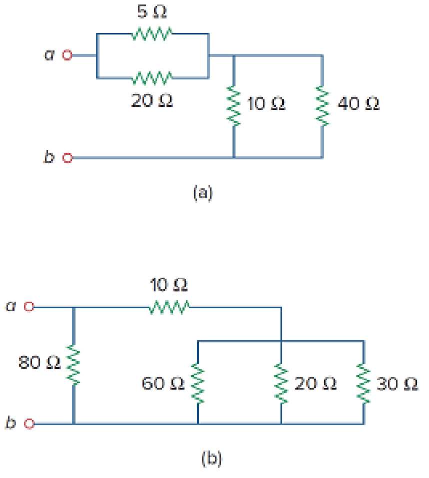
Chapter 2, Problem 43P

Problem

Calculate the equivalent resistance *Rab* at terminals *a-b* for each of the circuits in Fig. 2.107.

Figure 2.107:



Step-by-step solution

Step 1 of 17

(a)

Draw the following circuit diagram:

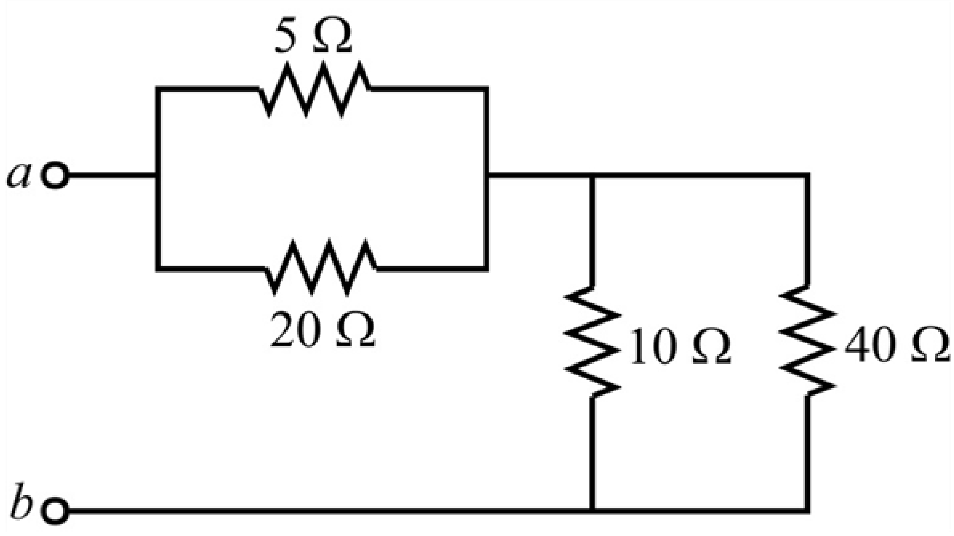


Figure 1

Step 2 of 17

Clearly from Figure 1, $20\ \Omega, 5\ \Omega$ and $10\ \Omega, 40\ \Omega$ are in parallel respectively.

Calculate the value of the parallel equivalent resistance R_{eq1} for $20\ \Omega, 5\ \Omega$.

Write the expression for the parallel equivalent resistance R_{eq1}.

$$\begin{aligned} R_{eq1} &= 20\ \Omega \,||\, 5\ \Omega \\ &= \frac{20\ \Omega(5\ \Omega)}{20\ \Omega + 5\ \Omega} \\ &= \frac{100}{25}\ \Omega \\ &= 4\ \Omega \end{aligned}$$

Therefore, the value of R_{eq1} is $4\ \Omega$.

Step 3 of 17

Calculate the value of the parallel equivalent resistance R_{eq2} for $10\ \Omega, 40\ \Omega$.

Write the expression for the parallel equivalent resistance R_{eq2}.

$$\begin{aligned} R_{eq2} &= 10\ \Omega \,||\, 40\ \Omega \\ &= \frac{10\ \Omega(40\ \Omega)}{10\ \Omega + 40\ \Omega} \\ &= \frac{400}{50}\ \Omega \\ &= 8\ \Omega \end{aligned}$$

Therefore, the value of R_{eq2} is $8\ \Omega$.

Step 4 of 17

Draw the modified Figure 1 as shown in Figure 2.

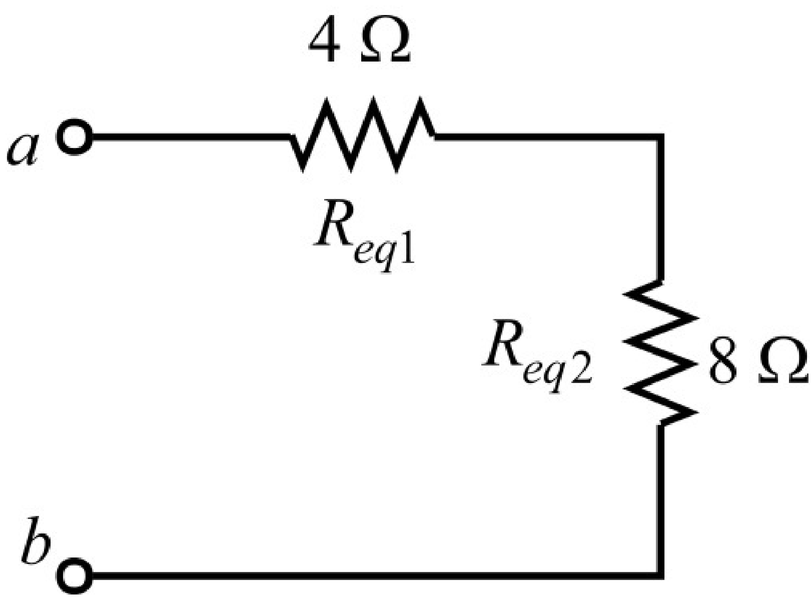


Figure 2

Step 5 of 17

Clearly from Figure 1, $4\ \Omega, 8\ \Omega$ are in series respectively.

Calculate the value of the series equivalent resistance R_{ab} for $4\ \Omega, 8\ \Omega$.

Write the expression for the series equivalent resistance R_{ab} for $4\ \Omega, 8\ \Omega$.

$$\begin{aligned} R_{ab} &= 4\ \Omega + 8\ \Omega \\ &= 12\ \Omega \end{aligned}$$

Therefore, the value of the series equivalent resistance R_{ab} for $4\ \Omega, 8\ \Omega$ is $12\ \Omega$.

Step 6 of 17

Draw the modified Figure 2 as shown in Figure 3.

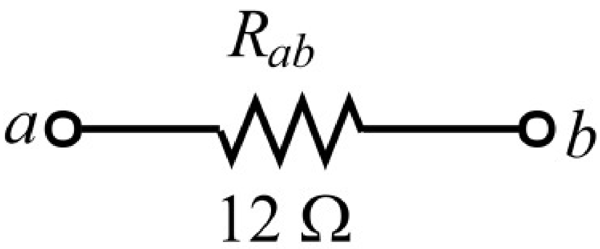


Figure 3

Step 7 of 17

Clearly from Figure 3, the value of required equivalent resistance R_{ab} is $\boxed{12\ \Omega}$.

Step 8 of 17

(b)

Draw the following circuit diagram:

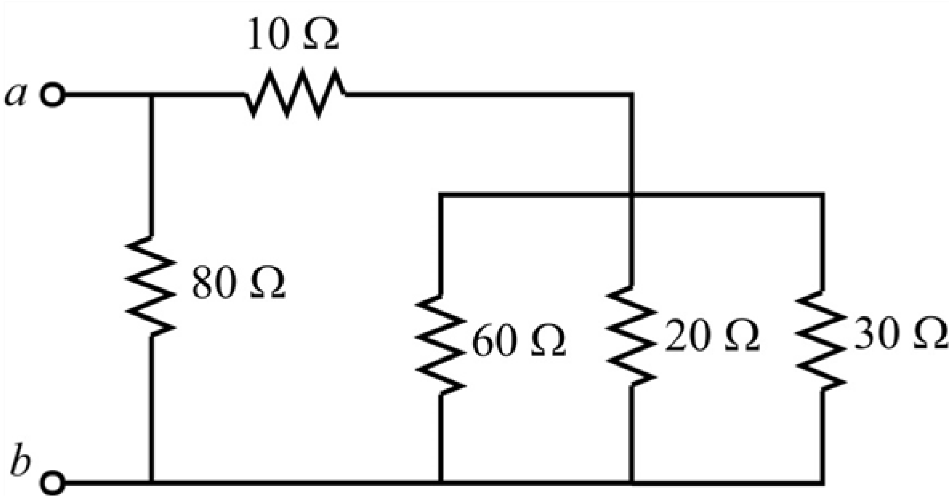


Figure 4

Step 9 of 17

Clearly from Figure 4, $20\ \Omega, 30\ \Omega, 60\ \Omega$ are in parallel.

Calculate the value of the parallel equivalent resistance R_{eq1} for $20\ \Omega, 30\ \Omega$.

Write the expression for the parallel equivalent resistance R_{eq1}.

$$\begin{aligned} R_{eq1} &= 20\ \Omega \,\|\, 30\ \Omega \\ &= \frac{20\ \Omega(30\ \Omega)}{20\ \Omega + 30\ \Omega} \\ &= \frac{600}{50}\ \Omega \\ &= 12\ \Omega \end{aligned}$$

Therefore, the value of R_{eq1} is $12\ \Omega$.

Step 10 of 17

Redraw the modified Figure 4 as shown in Figure 5.

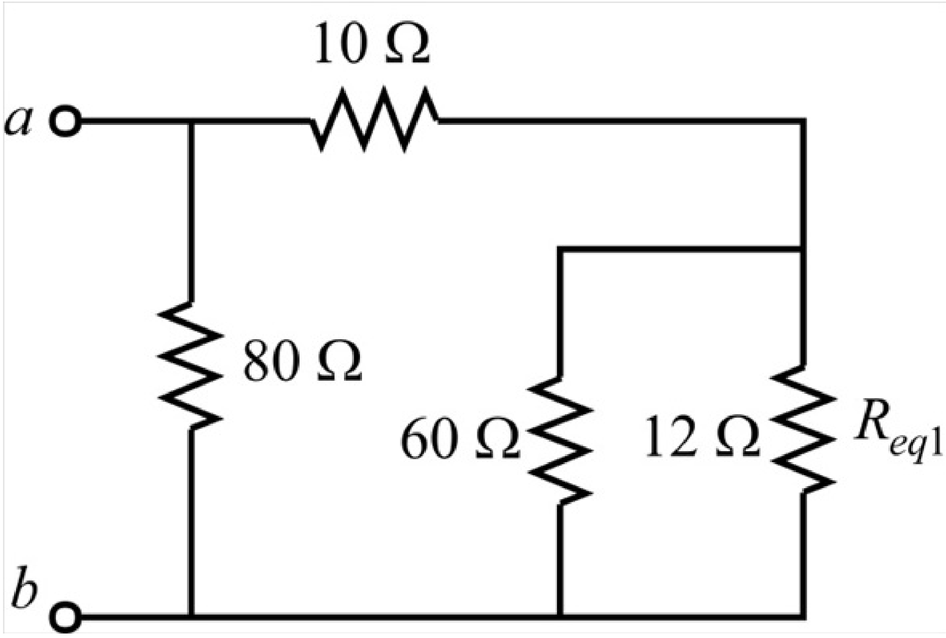


Figure 5

Step 11 of 17

Calculate the value of the parallel equivalent resistance R_{eq2} for $12\ \Omega, 60\ \Omega$.

Write the expression for the parallel equivalent resistance R_{eq2}.

$$\begin{aligned} R_{eq2} &= 12\ \Omega \,\|\, 60\ \Omega \\ &= \frac{12\ \Omega(60\ \Omega)}{12\ \Omega + 60\ \Omega} \\ &= \frac{720}{72}\ \Omega \\ &= 10\ \Omega \end{aligned}$$

Therefore, the value of R_{eq2} is $10\ \Omega$.

Step 12 of 17

Redraw the modified Figure 5 as shown in Figure 6.

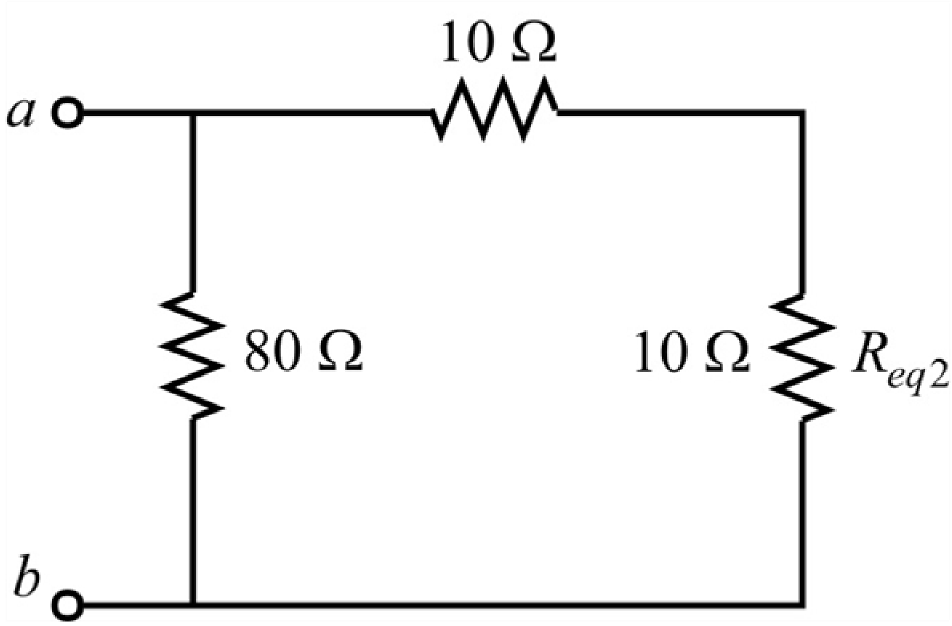


Figure 7

Step 13 of 17

Clearly from Figure 7, $10\ \Omega, 10\ \Omega$ are in series respectively.

Calculate the value of the series equivalent resistance R_{eq3} for $10\ \Omega, 10\ \Omega$.

Write the expression for the series equivalent resistance R_{eq3} for $10\ \Omega, 10\ \Omega$.

$$\begin{aligned} R_{ab} &= 10\ \Omega + 10\ \Omega \\ &= 20\ \Omega \end{aligned}$$

Therefore, the value of the series equivalent resistance R_{eq3} for $10\ \Omega, 10\ \Omega$ is $20\ \Omega$.

Step 14 of 17

Redraw the modified Figure 7 as shown in Figure 8.

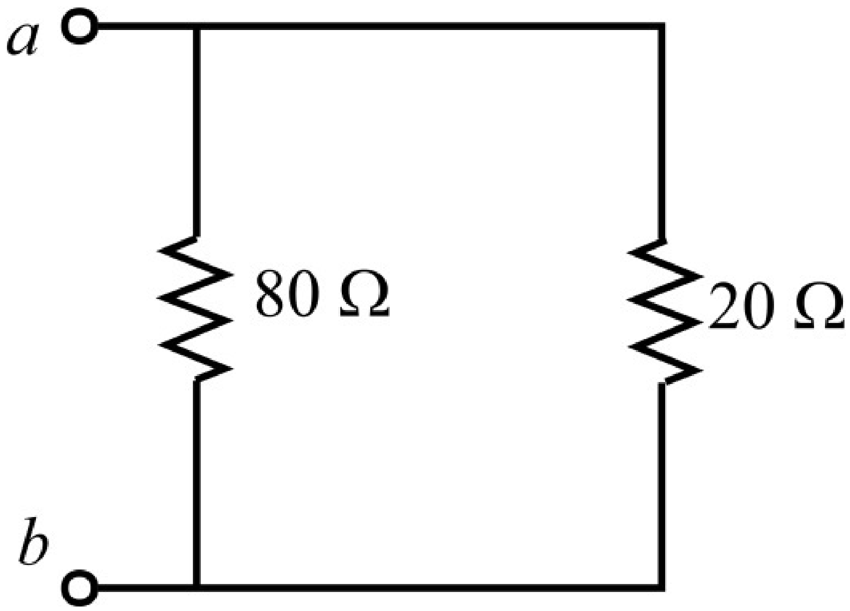


Figure 8

Step 15 of 17

Calculate the value of the parallel equivalent resistance R_{ab} for $80\ \Omega, 20\ \Omega$.

Write the expression for the parallel equivalent resistance R_{ab}.

$$\begin{aligned} R_{ab} &= 80\ \Omega \,||\, 20\ \Omega \\ &= \frac{80\ \Omega(20\ \Omega)}{80\ \Omega + 20\ \Omega} \\ &= \frac{1600}{100}\ \Omega \\ &= 16\ \Omega \end{aligned}$$

Therefore, the value of R_{ab} is $16\ \Omega$.

Step 16 of 17

Draw the modified Figure 8 as shown in Figure 9.

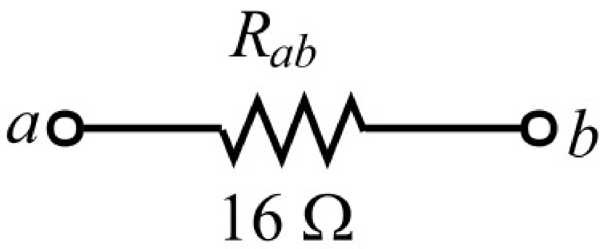


Figure 9

Step 17 of 17

Clearly from Figure 9, the value of required equivalent resistance R_{ab} is $\boxed{16\ \Omega}$.